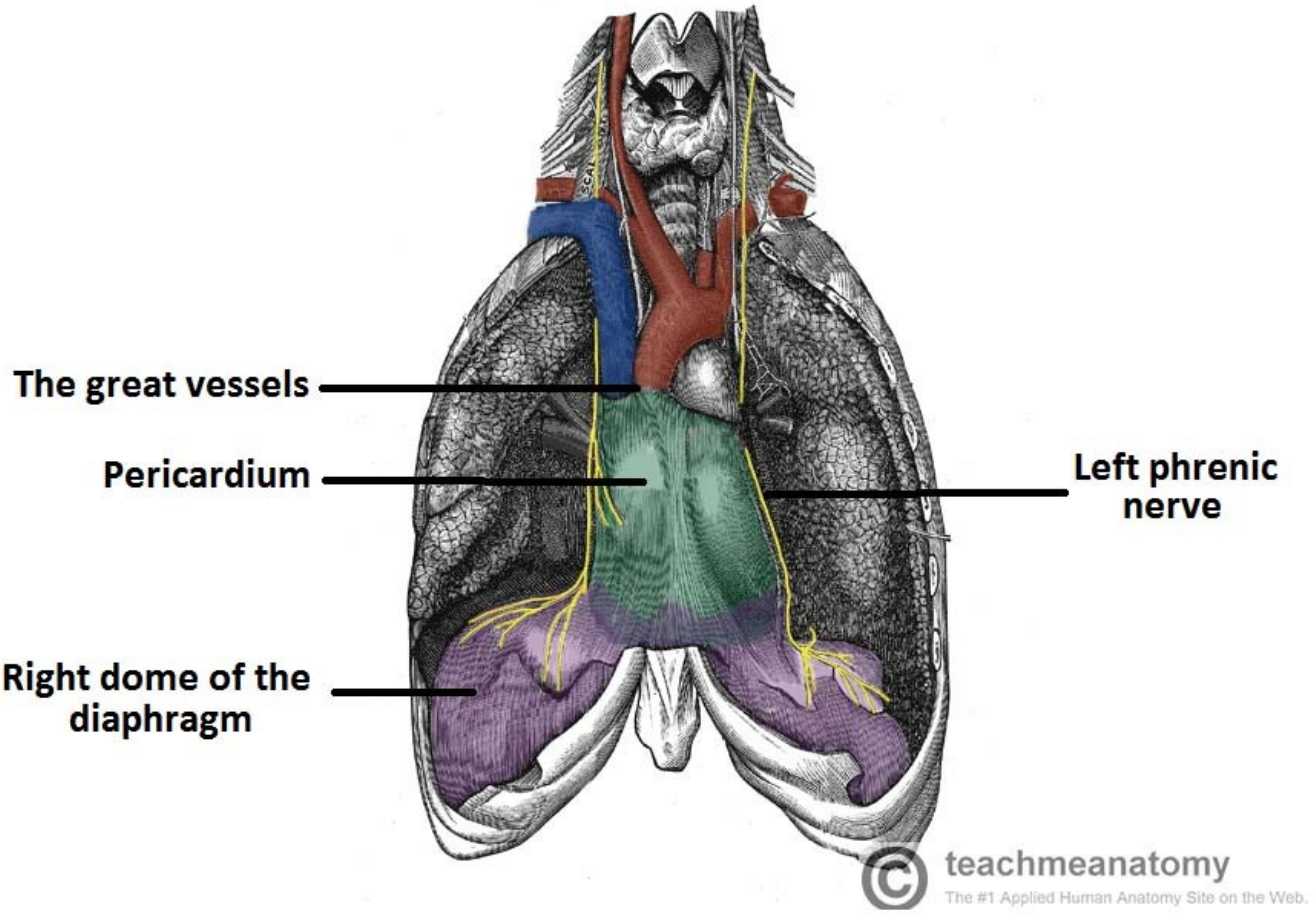
HEART

EXTERNAL FEATURES
INTERNAL FEATURES
GREAT VESSELS
CONDUCTING SYSTEM
PERICARDIUM
BLOOD SUPPLY

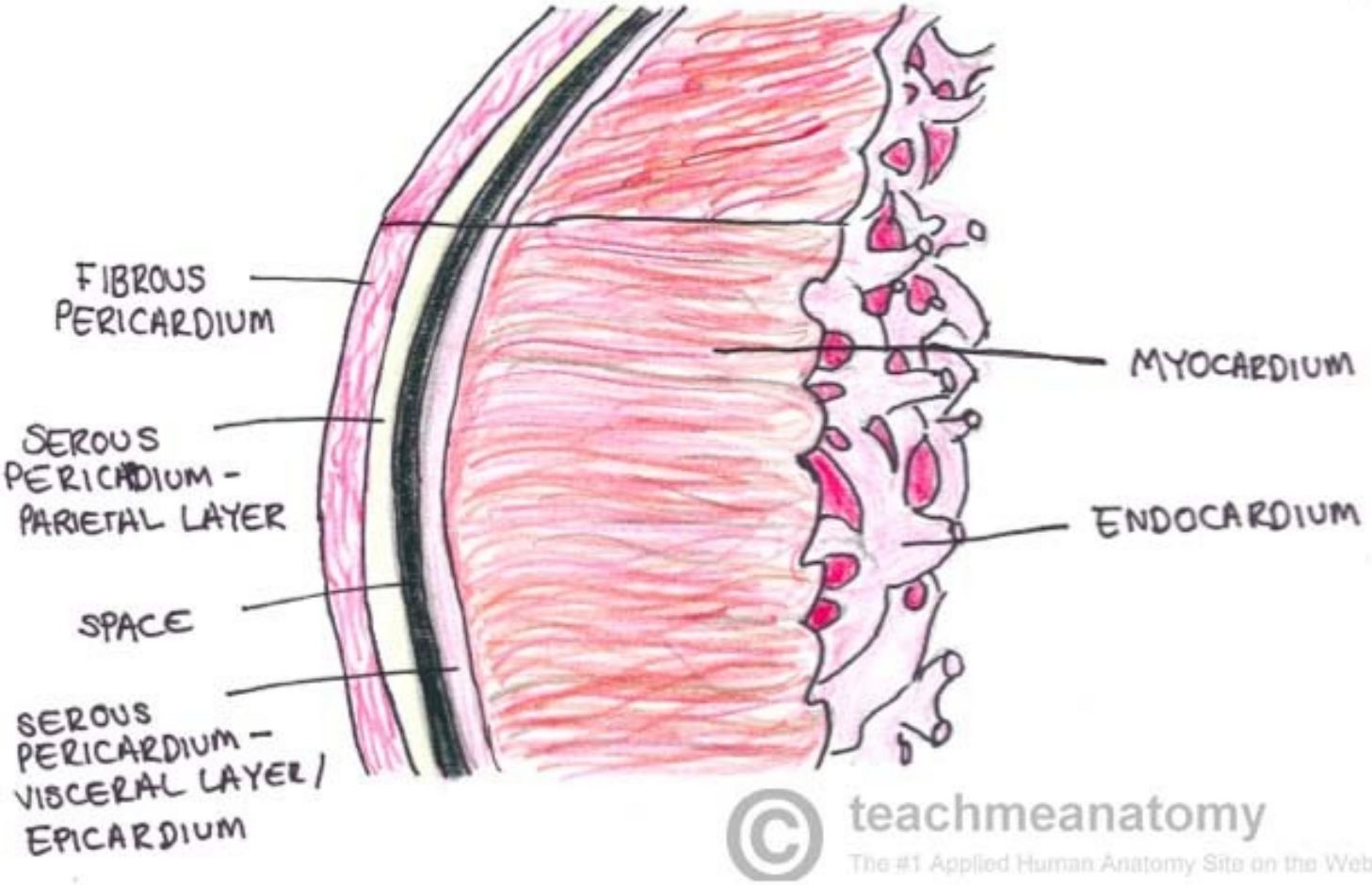
PERICARDIUM

- the pericardium is a **fibroserous**, fluid-filled sack that surrounds the muscular body of the heart and the roots of the great vessels (the aorta, pulmonary artery, pulmonary veins, and the superior and inferior vena cavae).
- The pericardium is made up of two main layers: a tough external layer known as the **fibrous pericardium,** and a thin, internal layer known as the **serous pericardium**


The great vessels
Pericardium
Right dome of the diaphragm
Left phrenic nerve
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- **<u>Fibrous Pericardium</u>**
- Continuous with the central tendon of the diaphragm, the fibrous pericardium is made of tough connective tissue and is relatively non-distensible. Its rigid structure prevents rapid overfilling of the heart

- **Serous Pericardium**
- Enclosed within the fibrous pericardium, the serous pericardium is itself divided into two layers: the **outer parietal layer** that lines the internal surface of the fibrous pericardium and the **internal visceral layer** that forms the outer layer of the heart (also known as the **epicardium**). Each layer is made up of a single sheet of epithelial cells, known as mesothelium.
- Found between the outer and inner serous layers is the **pericardial cavity**, which contains a small amount of lubricating serous fluid. The serous fluid serves to minimize the friction generated by the heart as it contracts.



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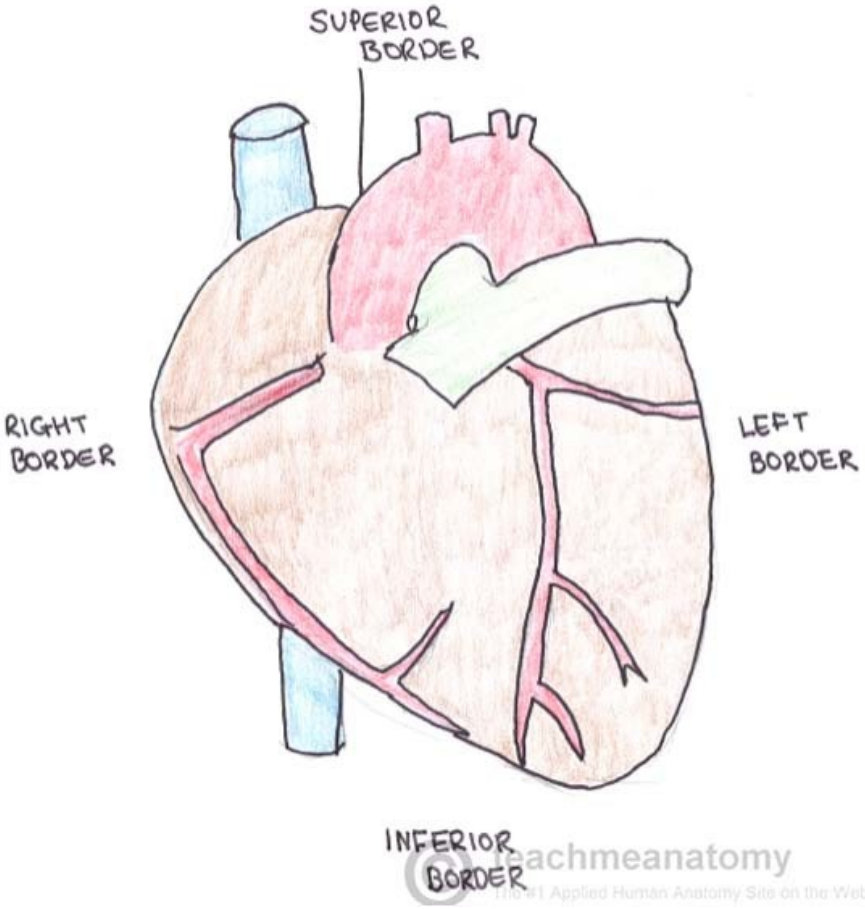
- **Innervation**
- The **phrenic nerve** (C3-C5) is responsible for the somatic innervation of the pericardium, as well as providing motor and sensory innervation to the diaphragm. Originating in the neck and travelling down through the thoracic cavity, the phrenic nerve is a common source of **referred pain,** with a key example being shoulder pain experienced as a result of pericarditis.

EXTERNAL FEATURES OF HEART

- The heart is a hollow muscular pump, which lies in the middle mediastinum. On its surface, it has several distinctive features which are of anatomical and clinical importance.

- **Orientation and Surfaces**
- The heart has been described by many texts as “a pyramid which has fallen over”. The **apex**of this pyramid pointing in an anterior-inferior direction.
- In its typical anatomical orientation, the heart has 5 surfaces, formed by different internal divisions of the heart:
- **Anterior (or sternocostal)** – Right ventricle.
- **Posterior (or base)** – Left atrium.
- **Inferior (or diaphragmatic)** – Left and right ventricles.
- **Right pulmonary** – Right atrium.
- **Left pulmonary** – Left ventricle.

- **Borders**
- Separating the surfaces of the heart are its borders. There are four main borders of the heart:
- **Right border** – Right atrium
- **Inferior border** – Left ventricle and right ventricle
- **Left border** – Left ventricle (and some of the left atrium)
- **Superior border** – Right and left atrium and the great vessels


SUPERIOR BORDER
RIGHT BORDER
LEFT BORDER
INFERIOR BORDER
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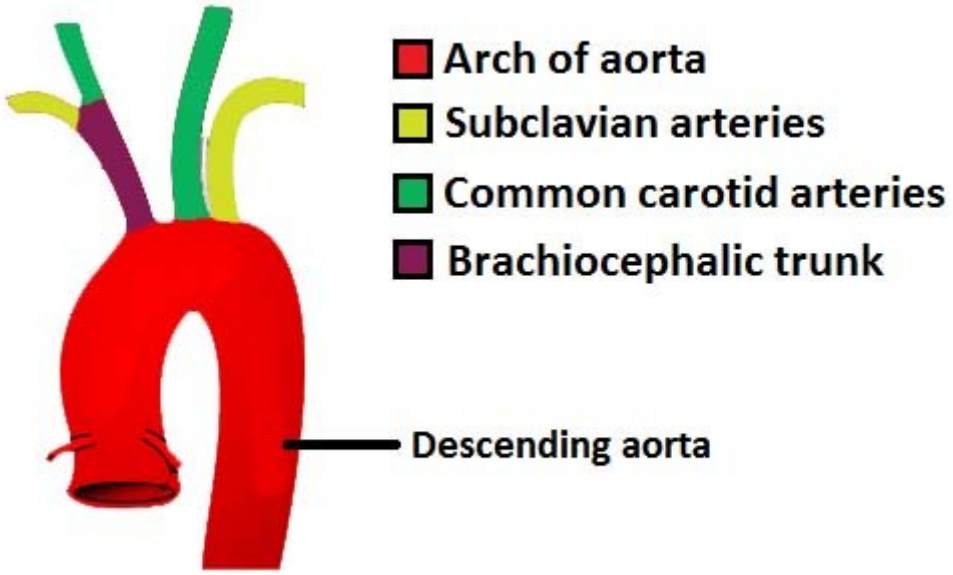
- **Sulci of the Heart**
- The heart is a hollow structure. On the interior, it is divided into four chambers. These divisions create grooves on the surface of the heart – these are known as sulci.
- The **coronary sulcus** (or atrioventricular groove) runs transversely around the heart – it represents the wall dividing the atria from the ventricles. The sinus contains important vasculature, such as the right coronary artery.
- The **anterior** and **posterior interventricular** sulci can be found running vertically on their respective sides of the heart. They represent the wall separating the ventricles.

- **Pericardial Sinuses**
- The pericardial sinuses are **not** the same as ‘anatomical sinuses’ (such as the paranasal sinuses). They are **passageways** formed the unique way in which the pericardium folds around the great vessels.
- The **oblique pericardial sinus** is a blind ending passageway (‘cul de sac’) located on the posterior surface of the heart.
- The **transverse pericardial sinus** is found superiorly on the heart. It can be used in coronary artery bypass grafting

GREAT VESSELS OF HEART

- The great vessels of the heart function to carry blood to and from the heart as it pumps, located largely within the middle mediastinum
  .

- **Aorta**
- The aorta is the largest artery in the body. It carries oxygenated blood (pumped by the left side of the heart) to the rest of the body.
- The aorta arises from the **aortic orifice** at the base of the left ventricle, with inflow via the **aortic valve**. Its first segment is known as the **ascending aorta**, which lies within the pericardium (covered by the visceral layer). From it branch the coronary arteries. The second continuous segment is the arch of the aorta, from which branch the major arteries to the head, neck and upper limbs. These are:
- **Brachiocephalic trunk**
- **Left common carotid artery**
- **Left subclavian artery**
- After the arch of the aorta, the aorta then becomes the **descending aorta** which continues down through the diaphragm into the abdomen.

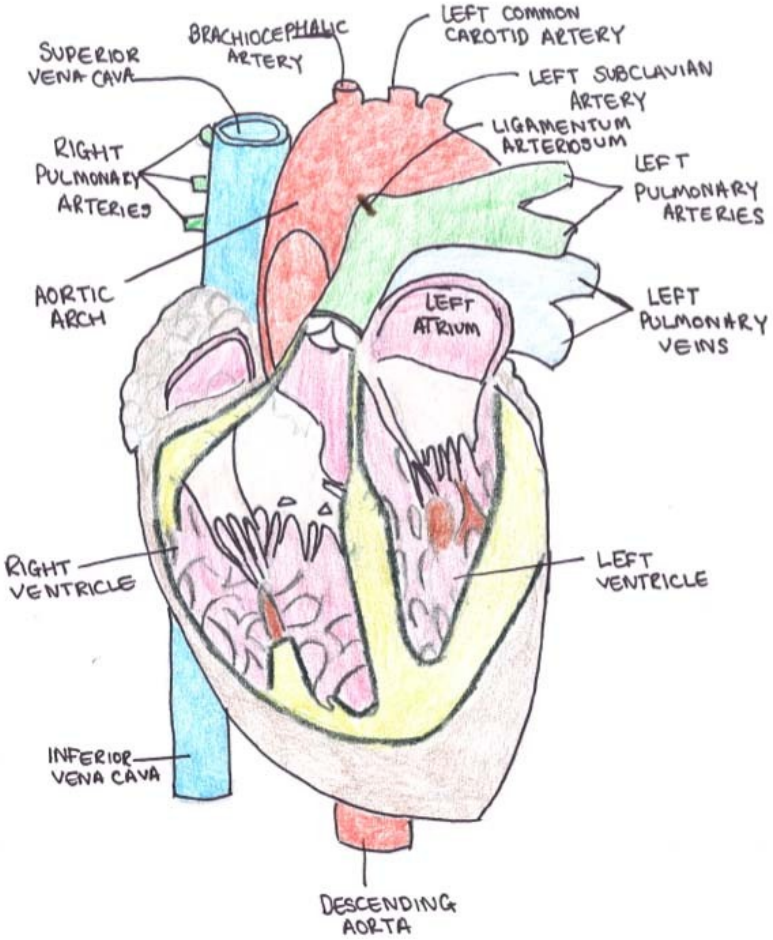

Arch of aorta
Subclavian arteries
Common carotid arteries
Brachiocephalic trunk
Descending aorta

- **Pulmonary Arteries**
- The pulmonary arteries receive **deoxygenated** blood from the right ventricle, and deliver it to the lungs for gas exchange to take place.
- The arteries begin as the pulmonary trunk, a thick and short vessel, which is separated from the right ventricle by the **pulmonary valve**. The trunk is located anteriorly and medially to the right atrium, sharing a common layer of pericardium with the ascending aorta. It continues upwards, overlapping the root of the aorta and passing posteriorly.
- At around the level of T5-T6, the **pulmonary** trunk splits into the right and left pulmonary arteries. The left pulmonary artery supplies blood to the left lung, bifurcating into two branches to supply each **lobe** of the lung. The right pulmonary artery is the thicker and longer artery of the two, supplying blood to the right lung. It also further divides into two branches.

- **Pulmonary Veins**
- The pulmonary veins receive **oxygenated**blood from the lungs, delivering it to the left side of the heart to be pumped back around the body.
- There are **four** pulmonary veins, with one superior and one inferior for each of the lungs. They enter the pericardium to drain into the superior left atrium, on the posterior surface. The **oblique pericardial sinus** can be found within the pericardium, between the left and right veins.
- The superior pulmonary veins return blood from the upper lobes of the lung, with the inferior veins returning blood from the lower lobes. The **inferior left pulmonary**vein is found at the hilum of the lung, while the **right inferior pulmonary** vein runs posteriorly to the superior vena cava and the right atrium.

- **Superior Vena Cava**
- The superior vena cava receives **deoxygenated** blood from the upper body (superior to the diaphragm, excluding the lungs and heart), delivering it to the right atrium.
- It is formed by merging of the **brachiocephalic veins**, travelling inferiorly through the thoracic region until draining into the superior portion of the right atrium at the level of the 3rd rib.
- As the superior vena cava makes its descent it is located in the right side of the superior mediastinum, before entering the middle mediastinum to lie beside the ascending aorta.

- **Inferior Vena Cava**
- The inferior vena cava receives **deoxygenated** blood from the lower body (all structures inferior to the diaphragm), delivering it back to the heart.
- It is initially formed in the pelvis by the **common iliac veins** joining together. It travels through the abdomen, collecting blood from the hepatic, lumbar, gonadal, renal and phrenic veins. The inferior vena cava then passes through the diaphragm, entering the pericardium at the level of T8. It drains into the inferior portion of the right atrium.


SUPERIOR VENA CAVA
BRACHIOCEPHALIC ARTERY
LEFT COMMON CAROTID ARTERY
LEFT SUBCLAVIAN ARTERY
LIGAMENTUM ARTERIOSUM
RIGHT PULMONARY ARTERIES
LEFT PULMONARY ARTERIES
AORTIC ARCH
LEFT ATRIUM
LEFT PULMONARY VEINS
RIGHT VENTRICLE
LEFT VENTRICLE
INFERIOR VENA CAVA
DESCENDING AORTA

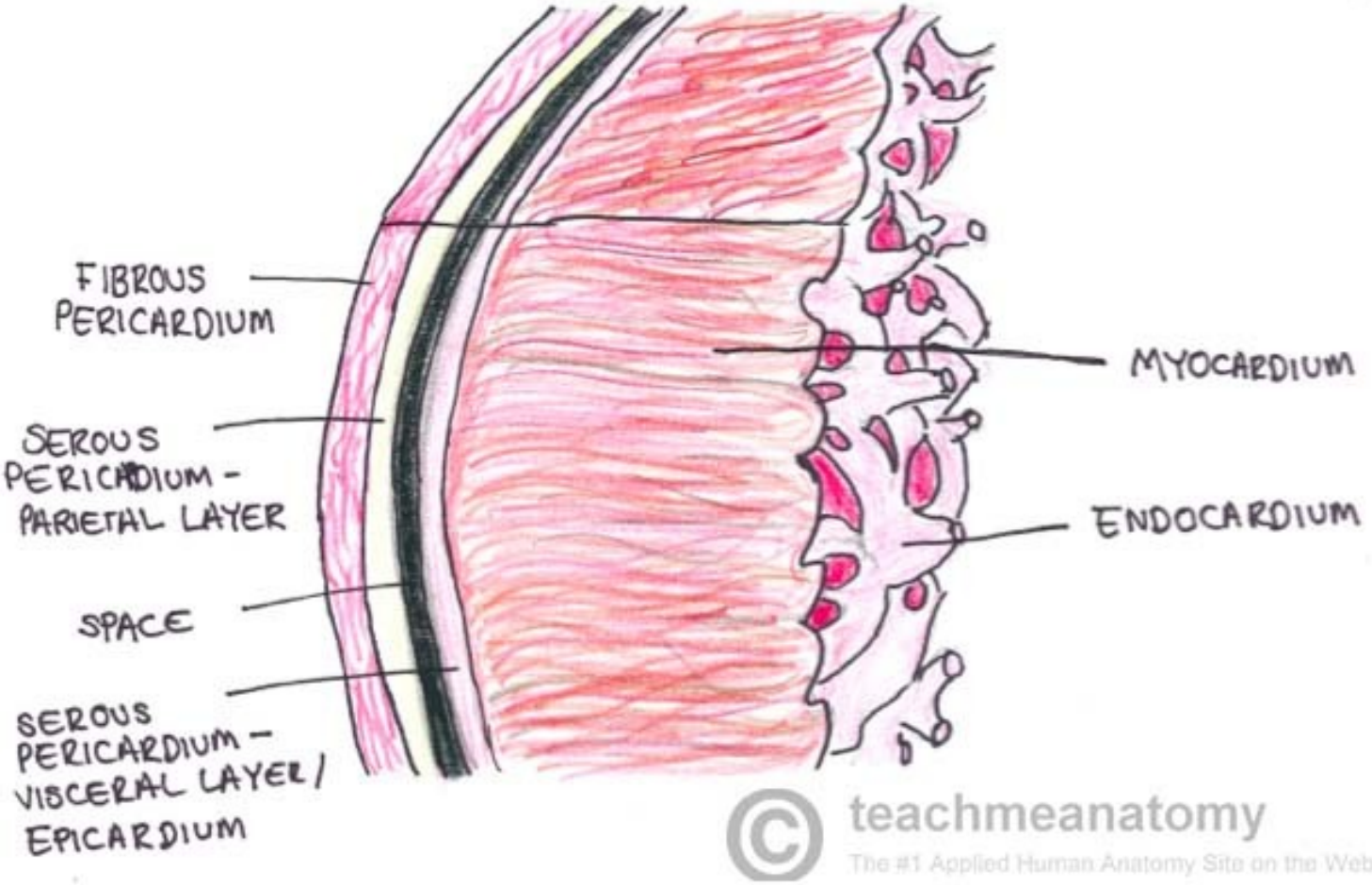
HEART WALL

- The heart wall itself can be divided into three distinct layers: the **endocardium, myocardium,** and **epicardium**.

- **Endocardium**
- The innermost layer of the cardiac wall is known as the endocardium. It lines the **cavities** and **valves** of the heart.
- Structurally, the endocardium is comprised of loose connective tissue and simple squamous epithelial tissue – it is similar in its composition to the endothelium which lines the inside of blood vessels.
- In addition to lining the inside of the heart, the endocardium also **regulates contractions** and aids cardiac **embryological development**.
- By **TeachMeSeries Ltd** (2018)
-

- **Subendocardial layer**
- The subendocardial layer lies between, and joins, the **endocardium** and the **myocardium**. It consists of a layer of loose fibrous tissue, containing the vessels and nerves of the conducting system of the heart. The **purkinje fibres** are located in this layer.
- **Myocardium**
- The myocardium is composed of cardiac muscle and is an involuntary striated muscle. The myocardium is responsible for contractions of the heart.

- **Subepicardial layer**
- The subepidcardial layer lies between, and joins, the **myocardium** and the **epicardium**.
- **Epicardium**
- The epicardium is the outermost layer of the heart, formed by the visceral layer of the pericardium. It is composed of connective tissue and fat. The connective tissue secretes a small amount of lubricating fluid into the **pericardial cavity**.
- In addition to the connective tissue and fat, the epicardium is lined by on its outer surface by simple squamous **epithelial** cells.



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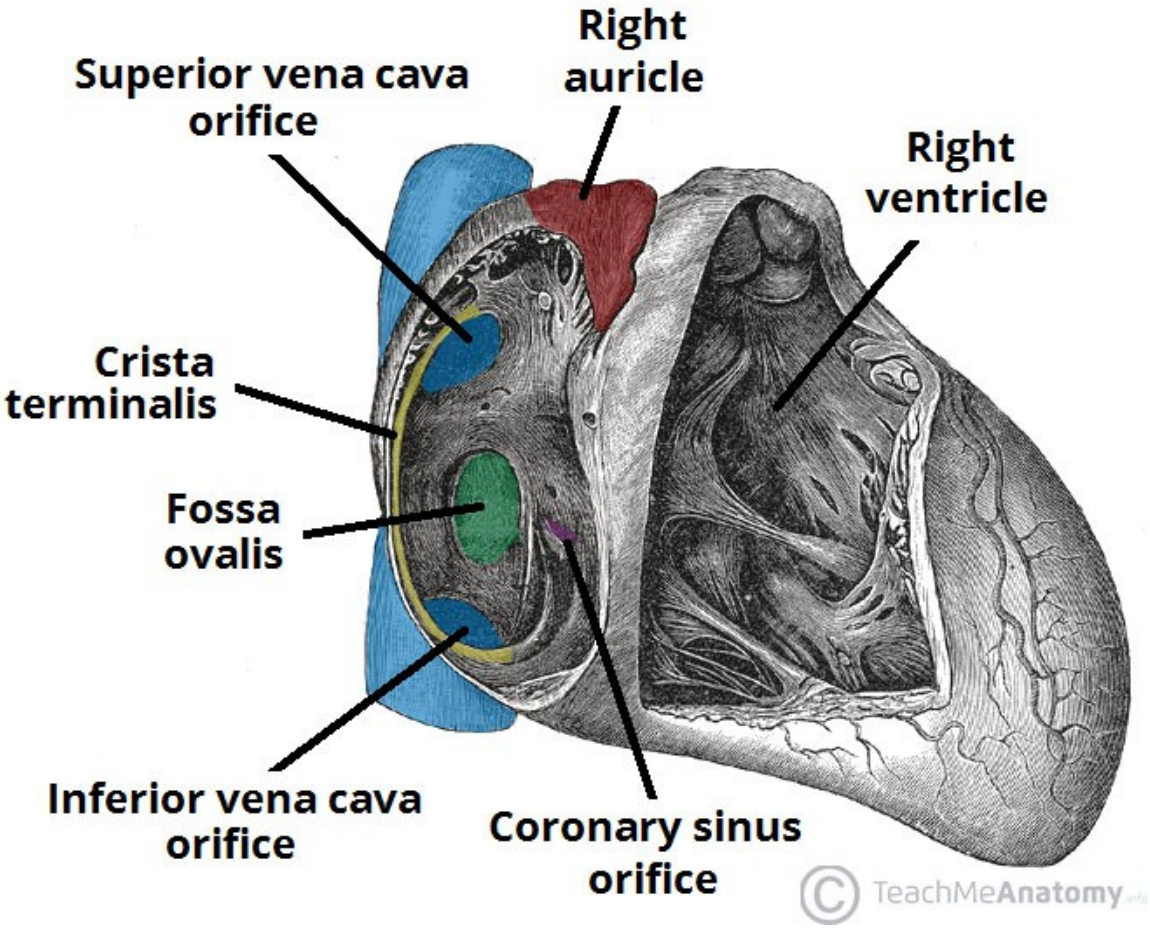
CHAMBERS OF HEART

- The heart consists of four chambers: the two atria and the **two ventricles**.
- Blood returning to the heart enters the atria, and is then pumped into the ventricles. From the left ventricle, blood passes into the aorta and enters the **systemic circulation**. From the right, it enters the **pulmonary circulation** via the pulmonary arteries.

- **Atria**
- **Right Atrium**
- The right atrium receives deoxygenated blood from the superior and inferior **vena cavae**, and from the coronary veins. It pumps this blood through the right **atrioventricular orifice**(guarded by the tricuspid valve) into the right ventricle.
- In the anatomical position, the right atrium forms the **right border** of the heart. Extending from the antero-medial portion of the chamber is the **right auricle** (right atrial appendage) – a muscular pouch that acts to increase the capacity of the atrium.

- The interior surface of the right atrium can be divided into two parts, each with a distinct embryological origin. These two parts are separated by a muscular ridge called the **crista terminalis**:
- **Sinus venarum** – located posterior to the crista terminalis. This part receives blood from the superior and inferior vena cavae. It has smooth walls and is derived from the embryonic sinus venosus.
- **Atrium proper** – located anterior to the crista terminalis, and includes the right auricle. It is derived from the primitive atrium, and has rough, muscular walls formed by pectinate muscles.
- The **coronary sinus** receives blood from the coronary veins. It opens into the right atrium between the inferior vena cava orifice and the right atrioventricular orifice.

- **Interatrial Septum**
- The **interatrial septum** is a solid muscular wall that separates the right and left atria.
- The septal wall in the right atrium is marked by a small oval-shaped depression called the **fossa ovalis**. This is the remnant of the **foramen ovale** in the foetal heart, which allows right to left shunting of blood to bypass the lungs. It closes once the newborn takes its first breath.

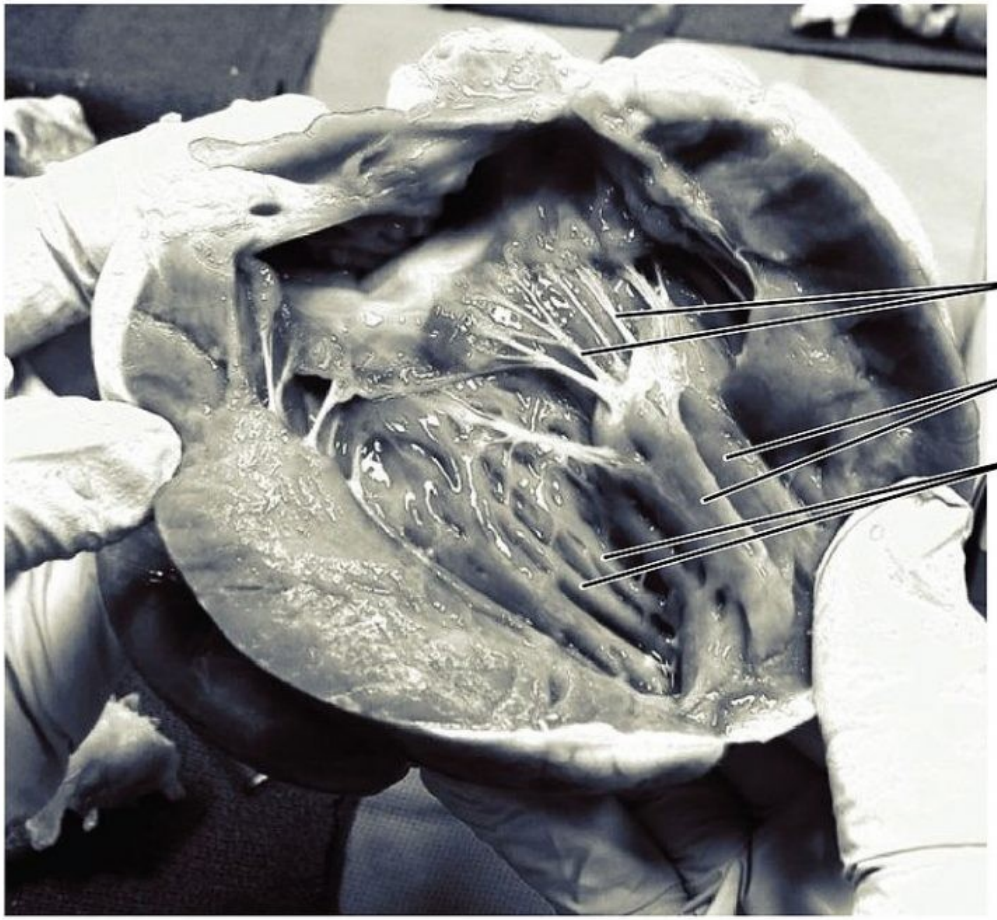

Superior vena cava orifice
Right auricle
Right ventricle
Crista terminalis
Fossa ovalis
Inferior vena cava orifice
Coronary sinus orifice
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- **Left Atrium**
- The left atrium receives oxygenated blood from the four **pulmonary veins**, and pumps it through the left atrioventricular orifice (guarded by the mitral valve) into the left ventricle.
- In the anatomical position, the left atrium forms the **posterior** border (base) of the heart. The**left auricle** extends from the superior aspect of the chamber, overlapping the root of the pulmonary trunk.
- The interior surface of the left atrium can be divided into two parts, each with a distinct embryological origin:
- **Inflow portion** – receives blood from the pulmonary veins. Its internal surface is smooth and it is derived from the pulmonary veins themselves.
- **Outflow portion** – located anteriorly, and includes the left auricle. It is lined by pectinate muscles, and is derived from the embryonic atrium.

- **Ventricles**
- The left and right ventricles of the heart receive blood from the atria and pump it into the outflow vessels; the **aorta** and the **pulmonary artery** respectively.
- **Right Ventricle**
- The right ventricle receives deoxygenated blood from the right atrium, and pumps it through the pulmonary orifice (guarded by the pulmonary valve), into the **pulmonary artery**.
- It is triangular in shape, and forms the majority of the **anterior** border of the heart. The right ventricle can be divided into an inflow and outflow portion, which are separated by a muscular ridge known as the **supraventricular crest**.

- **Inflow Portion**
- The interior of the inflow part of the right ventricle is covered by a series of irregular muscular elevations, called **trabeculae carnae**. They give the ventricle a ‘sponge-like’ appearance, and can be grouped into three main types:
- **Ridges** – attached along their entire length on one side to form ridges along the interior surface of the ventricle.
- **Bridges** – attached to the ventricle at both ends, but free in the middle. The most important example of this type is the moderator band, which spans between the interventricular septum and the anterior wall of the right ventricle. It has an important conductive function, containing the right bundle branches.
- **Pillars** (papillary muscles) – anchored by their base to the ventricles. Their apices are attached to fibrous cords (chordae tendineae), which are in turn attached to the three **tricuspid valve** cusps. By contracting, the papillary muscles ‘pull’ on the chordae tendineae to prevent prolapse of the valve leaflets during ventricular systole.

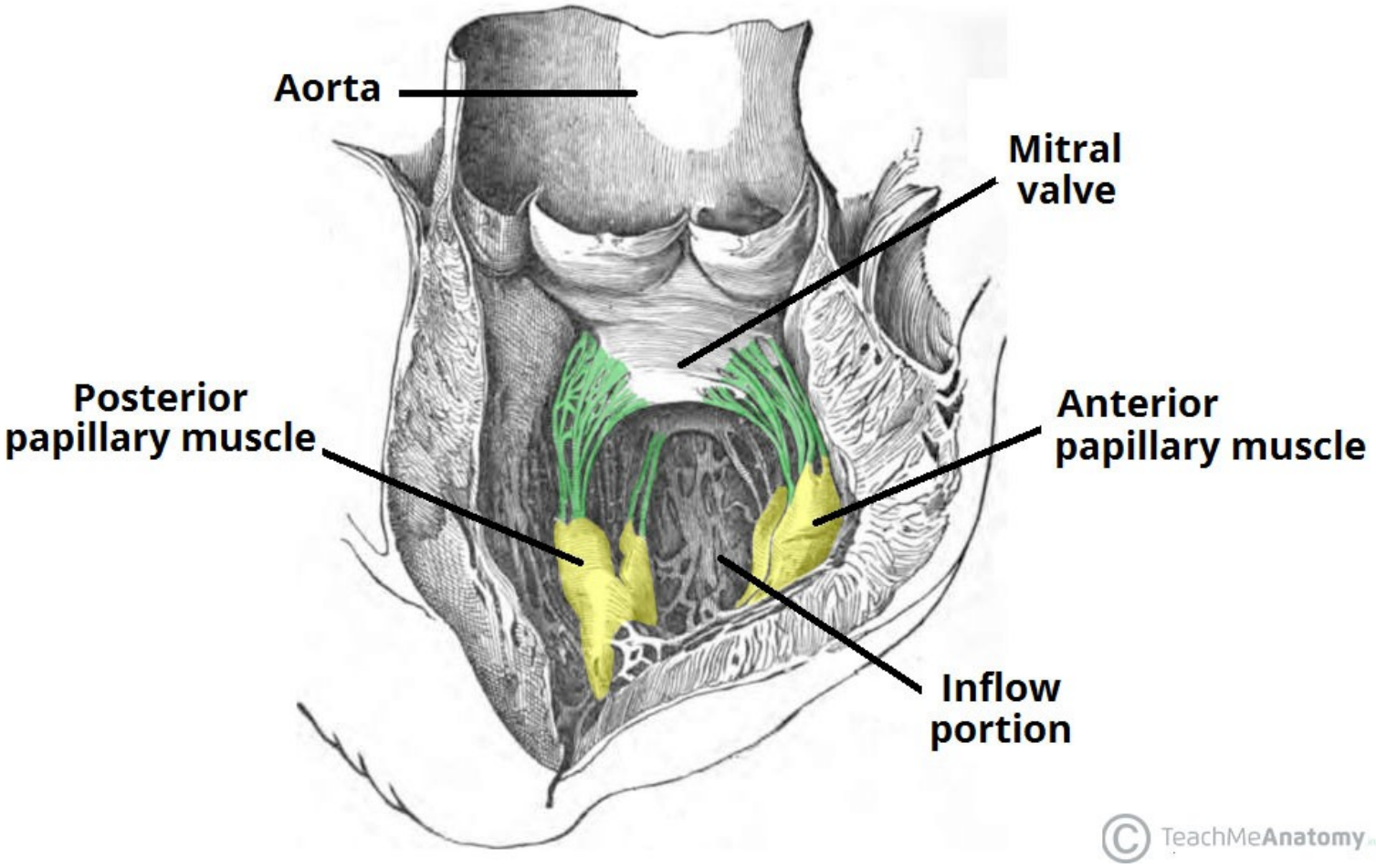
- **<u>Outflow Portion (Conus arteriosus)</u>**
- The outflow portion (leading to the pulmonary artery) is located in the superior aspect of the ventricle. It is derived from the embryonic **bulbus cordis**. It is visibly different from the rest of the right ventricle, with smooth walls and no trabeculae carneae.


Chordae tendineae
Papillary muscles
Trabeculae carnae
(ridges)

- **Interventricular Septum**
- The interventricular septum separates the two ventricles, and is composed of a superior **membranous** part and an inferior **muscular** part.
- The muscular part forms the majority of the septum and is the same thickness as the left ventricular wall. The membranous part is thinner, and part of the **fibrous skeleton** of the heart.

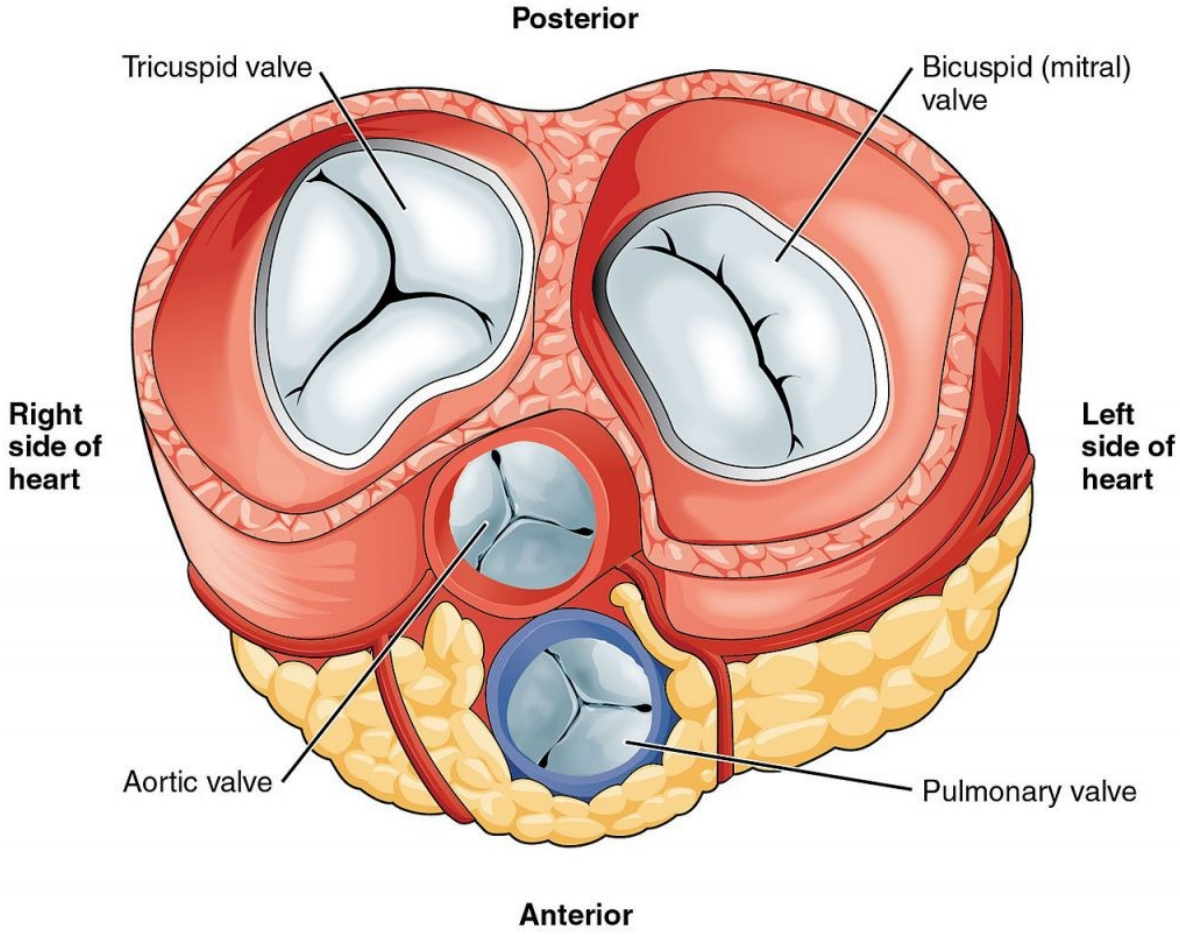
- **Left Ventricle**
- The left ventricle receives oxygenated blood from the left atrium, and pumps it through the aortic orifice (guarded by the aortic valve) into the **aorta**.
- In the anatomical position, the left ventricle forms the apex of the heart, as well as the left and diaphragmatic borders. Much like the right ventricle, it can be divided into an inflow portion and an outflow portion.

- **Inflow Portion**
- The walls of the inflow portion of the left ventricle are lined by **trabeculae carneae**, as described with the right ventricle. There are two papillary muscles present which attach to the cusps of the mitral valve.
- **Outflow Portion**
- The outflow part of the left ventricle is known as the **aortic vestibule**. It is smooth-walled with no trabeculae carneae, and is a derivative of the embryonic **bulbus cordis**.


Aorta
Mitral valve
Posterior papillary muscle
Anterior papillary muscle
Inflow portion
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VALVES OF HEART

- The valves of the heart are structures which ensure blood flows in only one direction. They are composed of connective tissue and endocardium (the inner layer of the heart).
- There are four valves of the heart, which are divided into two categories:
- **Atrioventricular valves**: The tricuspid valve and mitral (bicuspid) valve. They are located between the atria and corresponding ventricle.
- **Semilunar valves**: The pulmonary valve and aortic valve. They are located between the ventricles and their corresponding artery, and regulate the flow of blood leaving the heart.

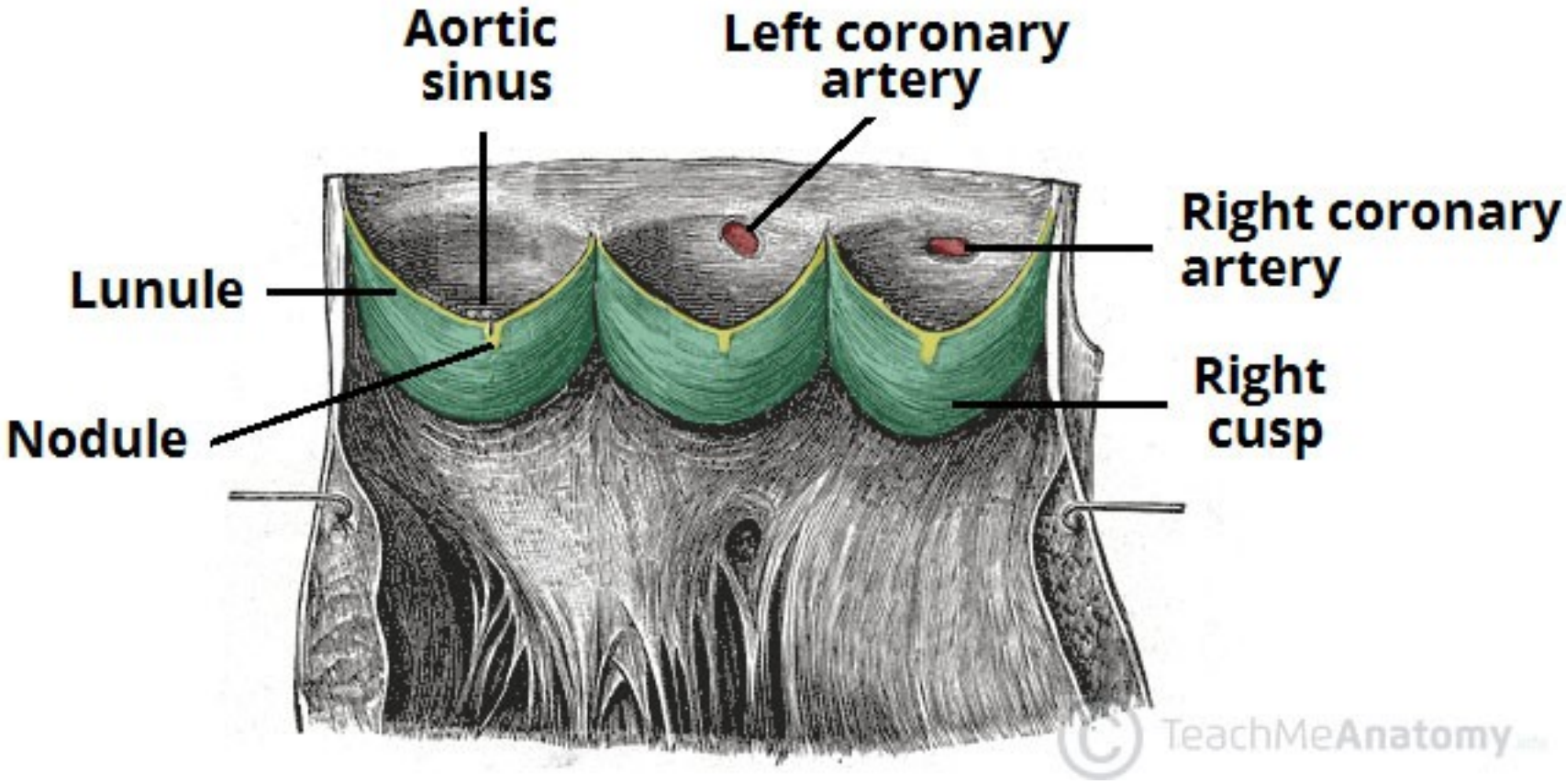

Posterior
Tricuspid valve
Bicuspid (mitral) valve
Right side of heart
Left side of heart
Aortic valve
Pulmonary valve
Anterior

- **Atrioventricular Valves**
- The atrioventricular valves are located between the atria and the ventricles. They close during the start of **ventricular contraction** (systole), producing the first heart sound. There are two AV valves:
- **Tricuspid valve** – located between the right atrium and the right ventricle (right atrioventricular orifice). It consists of three cusps (anterior, septal and posterior), with the base of each cusp anchored to a fibrous ring that surrounds the orifice.
- **Mitral valve** – located between the left atrium and the left ventricle (left atrioventricular orifice). It is also known as the bicuspid valve because it has two cusps (anterior and posterior). Like the tricuspid valve, the base of each cusp is secured to fibrous ring that surrounds the orifice.

- The mitral and tricuspid valves are supported by the attachment of fibrous cords (**chordae tendineae**) to the free edges of the valve cusps. The chordae tendineae are, in turn, attached to **papillary muscles**, located on the interior surface of the ventricles – these muscles contract during ventricular systole to prevent prolapse of the valve leaflets into the atria.
- There are **five** papillary muscles in total. Three are located in the right ventricle, and support the tricuspid valve. The remaining two are located within the left ventricle, and act on the mitral valve.

- **Semilunar Valves**
- The semilunar valves are located between the ventricles and outflow vessels. They close at the beginning of **ventricular relaxation** (diastole), producing the second heart sounds. There are two semilunar valves:
- **Pulmonary valve** – located between the right ventricle and the pulmonary trunk (pulmonary orifice). The valve consists of three cusps – left, right and anterior (named by their position in the foetus before the heart undergoes rotation).

- **Aortic valve** – located between the left ventricle and the ascending aorta (aortic orifice). The aortic valve consists of three cusps – right, left and posterior.
  - The left and right aortic sinuses mark the origin of the left and right coronary arteries. As blood recoils during ventricular diastole, it fills the aortic sinuses and enters the coronary arteries to supply the myocardium.
- The pulmonary and aortic valves have a similar structure. The sides of each valve leaflet are attached to the walls of the outflow vessel, which is slightly dilated to form a **sinus**. The free superior edge of each leaflet is thickened (the **lunule**), and is widest in the midline (the **nodule**).
- At the beginning of ventricular diastole, blood flows back towards the heart, filling the sinuses and pushing the valve cusps together. This closes the valve.


Aortic sinus
Left coronary artery
Right coronary artery
Lunule
Nodule
Right cusp
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CONDUCTING SYSTEM OF HEART

- The **cardiac conduction system** is a collection of nodes and specialised conduction cells that initiate and co-ordinate contraction of the heart muscle. It consists of:
- Sinoatrial node
- Atrioventricular node
- Atrioventricular bundle (bundle of His)
- Purkinje fibres

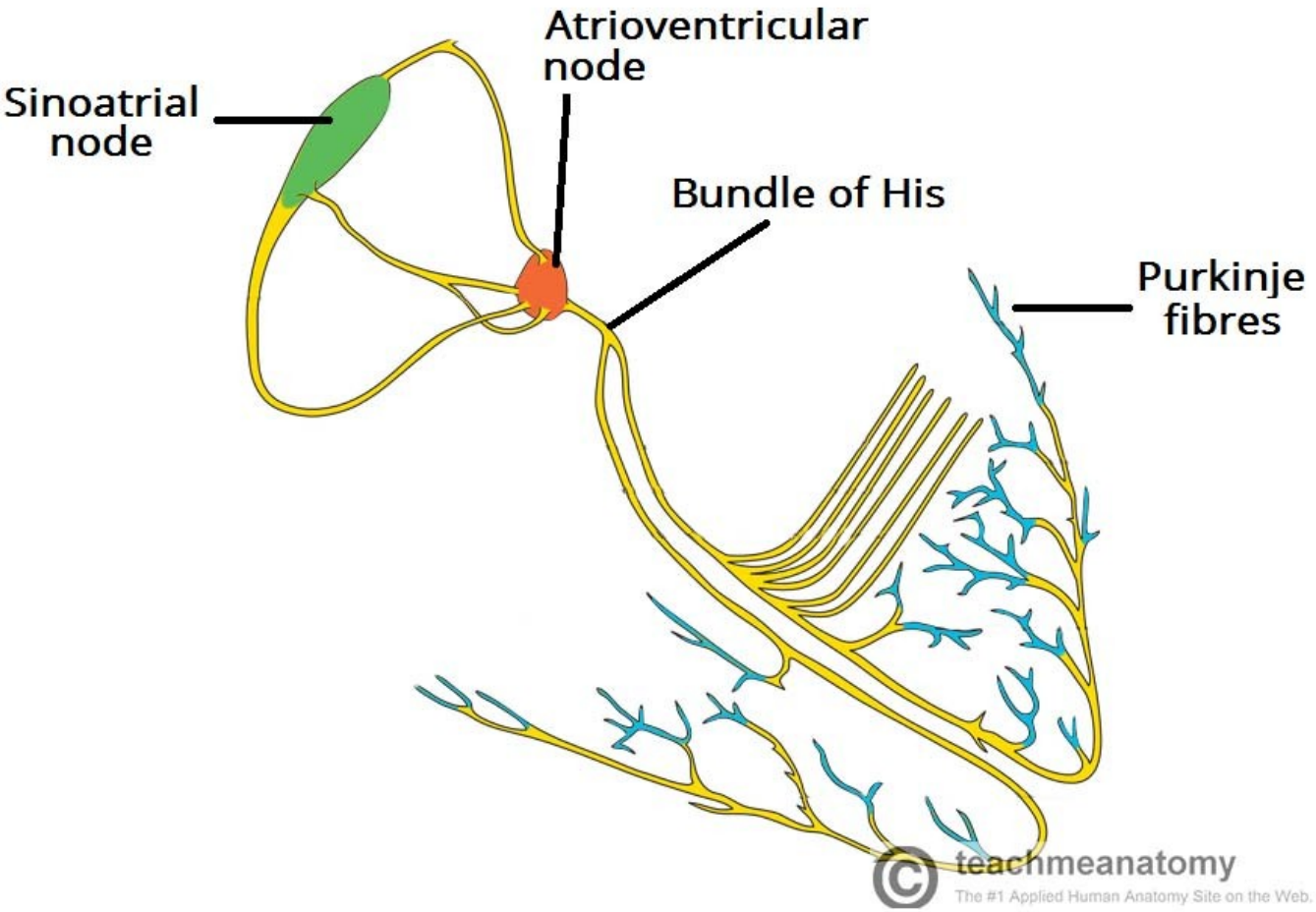
- The sequence of electrical events during one full contraction of the heart muscle:
- An excitation signal (an action potential) is created by the **sinoatrial (SA) node**.
- The wave of excitation spreads across the **atria,**causing them to contract.
- Upon reaching the **atrioventricular (AV) node**, the signal is delayed.
- It is then conducted into the **bundle of His**, down the interventricular septum.
- The bundle of His and the **Purkinje fibres** spread the wave impulses along the ventricles, causing them to contract.

- **Sinoatrial Node**
- The **sinoatrial (SA) node** is a collection of specialised cells (pacemaker cells), and is located in the upper wall of the right atrium, at the junction where the superior vena cava enters.
- These pacemaker cells can spontaneously generate **electrical impulses**. The wave of excitation created by the SA node spreads via gap junctions across both atria, resulting in atrial contraction (atrial systole) – with blood moving from the atria into the ventricles.
- The rate at which the SA node generates impulses is influenced by the autonomic nervous system:
- **Sympathetic nervous system** – increases firing rate of the SA node, and thus increases heart rate.
- **Parasympathetic nervous system** – decreases firing rate of the SA node, and thus decreases heart rate.

- **Atrioventricular Node**
- After the electrical impulses spread across the atria, they converge at the **atrioventricular node** – located within the atrioventricular septum, near the opening of the coronary sinus.
- The AV node acts to delay the impulses by approximately **120ms**, to ensure the atria have enough time to fully eject blood into the ventricles before ventricular systole.
- The wave of excitation then passes from the atrioventricular node into the atrioventricular bundle.

- **Atrioventricular Bundle**
- The **atrioventricular bundle** (bundle of His) is a continuation of the specialised tissue of the AV node, and serves to transmit the electrical impulse from the AV node to the Purkinje fibres of the ventricles.
- It descends down the membranous part of the interventricular septum, before dividing into two main bundles:
- **Right bundle branch** – conducts the impulse to the Purkinje fibres of the right ventricle
- **Left bundle branch** – conducts the impulse to the Purkinje fibres of the left ventricle.
-

- **Purkinje Fibres**
- The **Purkinje fibres** (sub-endocardial plexus of conduction cells) are a network of specialised cells. They are abundant with glycogen and have extensive gap junctions.
- These cells are located in the **subendocardial surface** of the ventricular walls, and are able to rapidly transmit cardiac action potentials from the atrioventricular bundle to the myocardium of the ventricles.
- This rapid conduction allows **coordinated ventricular contraction** (ventricular systole) and blood is moved from the right and left ventricles to the pulmonary artery and aorta respectively.

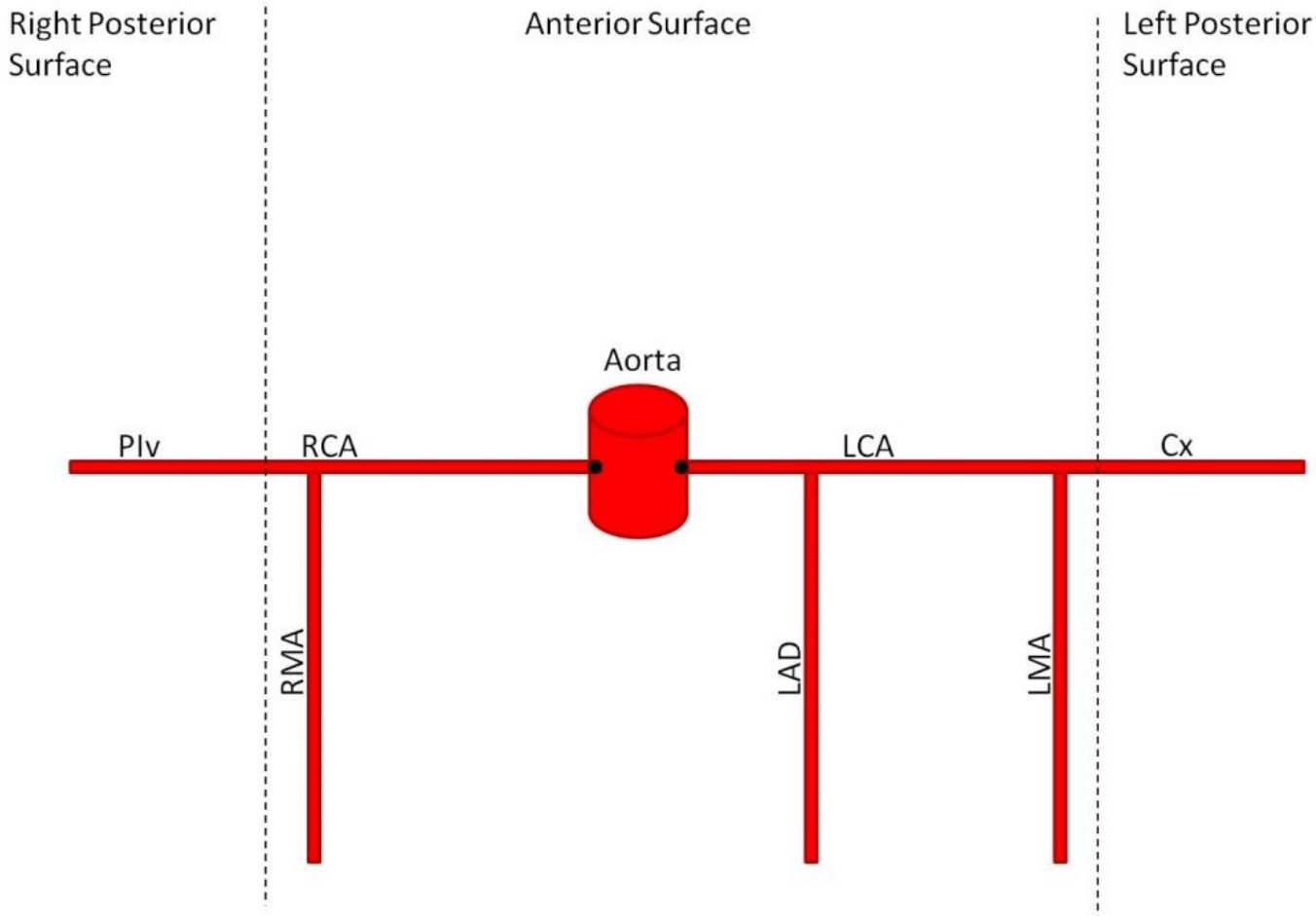

Sinoatrial node
Atrioventricular node
Bundle of His
Purkinje fibres
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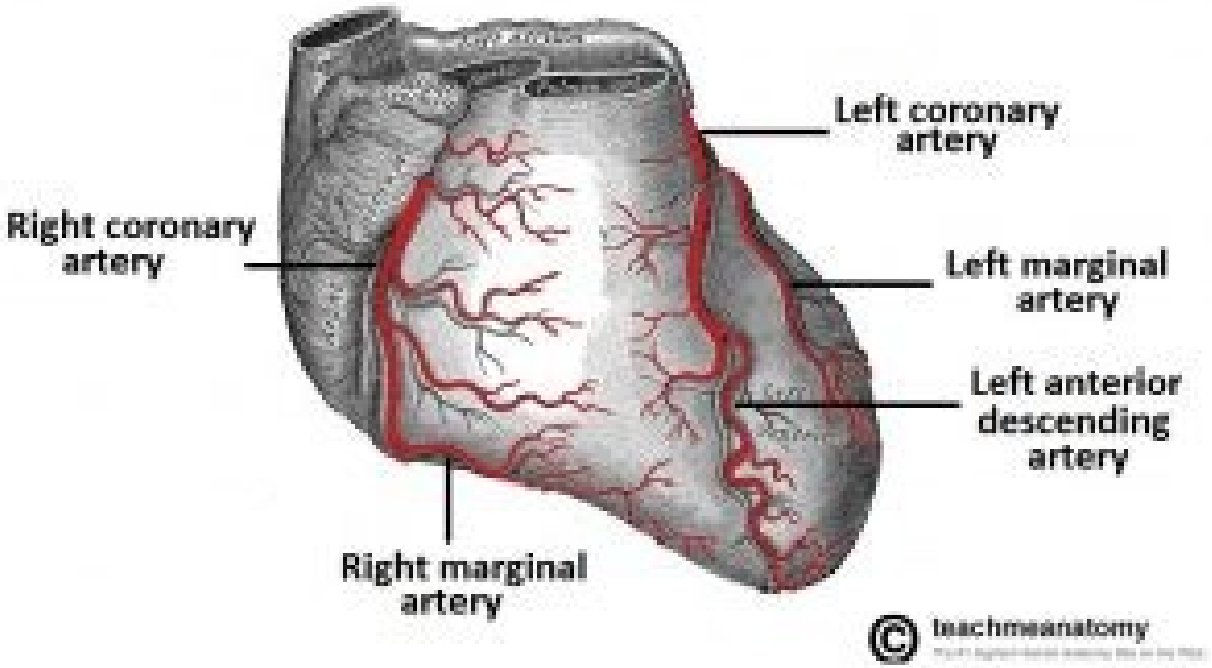
VASCULATURE OF HEART

- The entire body must be supplied with nutrients and oxygen via the **circulatory system** and the heart is no exception. The coronary circulation refers to the vessels that supply and drain the heart. Coronary arteries are named as such due to the way they encircle the heart, much like a crown.

- **Coronary Arteries**
- There are two main coronary arteries which branch to supply the entire heart. They are named the left and right coronary arteries, and arise from the left and right **aortic sinuses** within the aorta.
- The aortic sinuses are small openings found within the aorta behind the left and right flaps of the **aortic valve**. When the heart is relaxed, the back-flow of blood fills these valve pockets, therefore allowing blood to enter the coronary arteries.
- The **left coronary artery** (LCA) initially branches to yield the left anterior descending (LAD), also called the anterior interventricular artery. The LCA also gives off the left marginal artery (LMA) and the left circumflex artery (Cx). In ~20-25% of individuals, the left circumflex artery contributes to the posterior interventricular artery (PIv).
- The **right coronary artery** (RCA) branches to form the right marginal artery (RMA) anteriorly. In 80-85% of individuals, it also branches into the posterior interventricular artery (PIv) posteriorly.

- **Distribution of the Coronary Arteries**
- In general, the area of the heart which an artery passes over will be the area that it perfuses. The following describes the anatomical course of the coronary arteries.

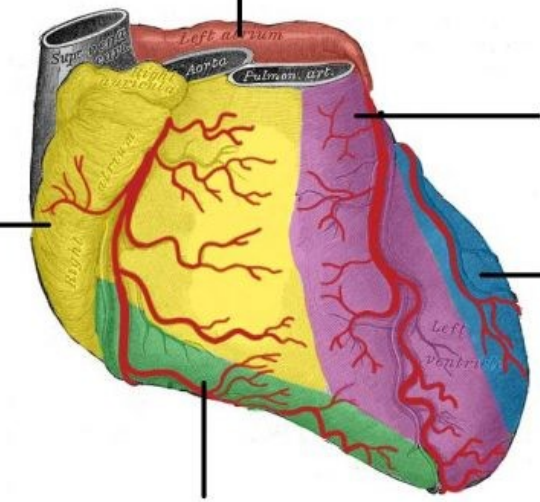

Right Posterior Surface
Anterior Surface
Left Posterior Surface
Aorta
PIv
RCA
LCA
Cx
RMA
LAD
LMA


Left coronary artery
Right coronary artery
Left marginal artery
Left anterior descending artery
Right marginal artery
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- The RCA passes to the right of the **pulmonary trunk** and runs along the coronary sulcus before branching. The right marginal artery arises from the RCA and moves along the right and inferior border of the heart towards the **apex**. The RCA continues to the posterior surface of the heart, still running along the **coronary sulcus**. The posterior interventricular artery then arises from the RCA and follows the posterior interventricular groove towards the apex of the heart.M

- The LCA passes between the left side of the pulmonary trunk and the **left auricle**. The LCA divides into the anterior interventricular branch and the circumflex branch. The anterior interventricular branch (LAD) follows the **anterior interventricular groove** towards the apex of the heart where it continues on the posterior surface to anastomose with the posterior interventricular branch. The **circumflex branch** follows the coronary sulcus to the left border and onto the posterior surface of the heart. This gives rise to the left marginal branch which follows the left border of the heart.

Left Circumflex Artery
Supplies the left atrium and left ventricle



Left Anterior Descending Artery
Supplies the right ventricle, left ventricle and interventricular septum

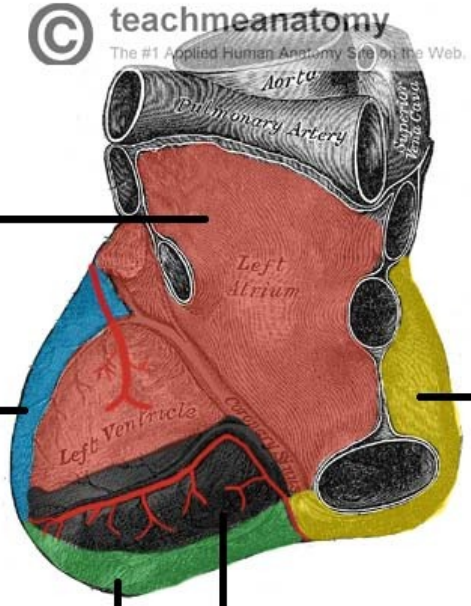
Right Coronary Artery
Supplies the right atrium and the right ventricle

Left Marginal Artery
Supplies the left ventricle

Right Marginal Artery
Supplies the right ventricle and the apex

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Left Circumflex Artery
Supplies the left atrium and ventricle

Left Marginal Artery
Supplies the left venticle

Right Coronary Artery
Supplies the right atrium and right ventricle

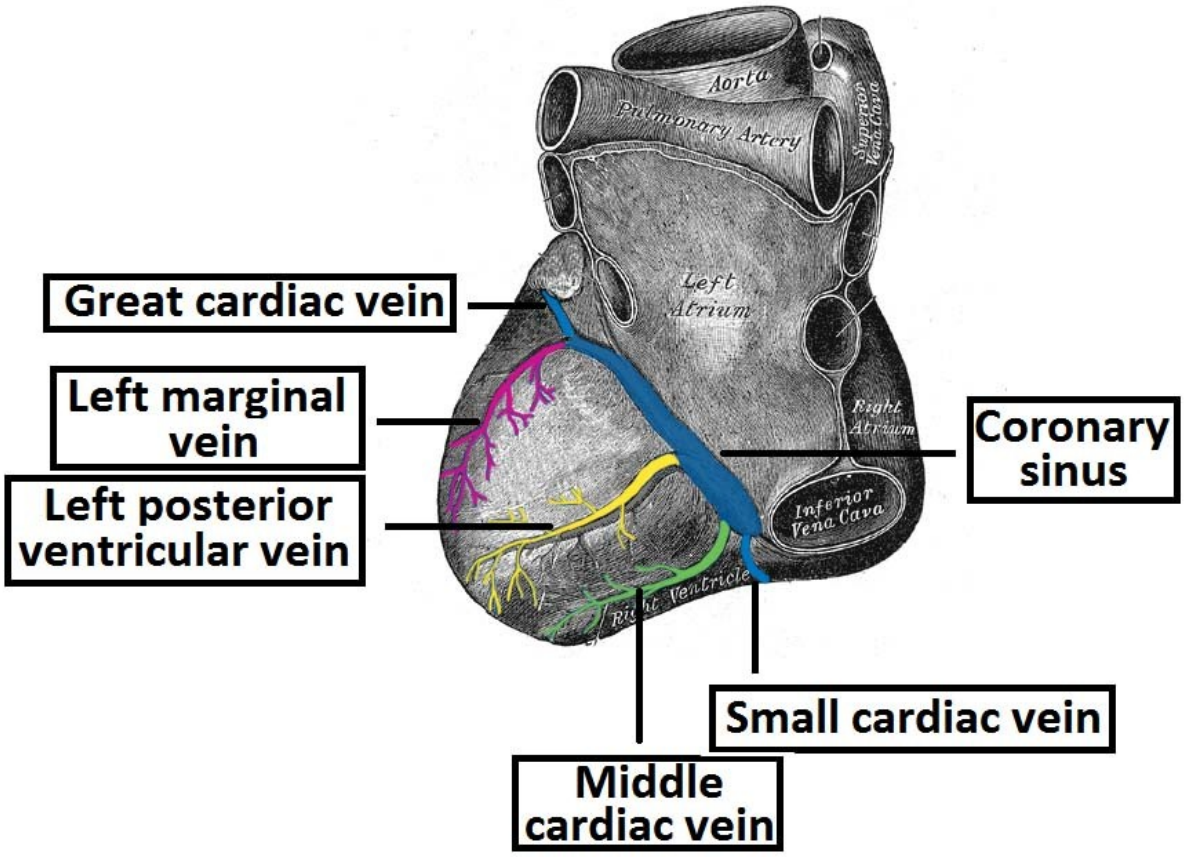
Right Marginal Artery
Supplies the right ventricle and apex

Posterior Interventricular Artery
Supplies the right and left ventricles, and the interventricular septum

- **Cardiac Veins**
- Blood travels from the subendocardium into the **thebesian veins**, which are small tributaries running throughout the myocardium. These in turn drain into larger veins that empty into the **coronary sinus**. The coronary sinus is the main vein of the heart, located on the posterior surface in the **coronary sulcus**, which runs between the left atrium and left ventricle. The sinus drains into the **right atrium**. Within the right atrium, the opening of the coronary sinus is located between the right atrioventricular orifice and the inferior vena cava orifice.

- There are **five tributaries** which drain into the coronary sinus:
- The **great cardiac vein** is the main tributary. It originates at the apex of the heart and follows the anterior interventricular groove into the coronary sulcus and around the left side of the heart to join the coronary sinus.
- The **small cardiac vein** is also located on the anterior surface of the heart. This passes around the right side of the heart to join the coronary sinus.
- Another vein which drains the right side of the heart is the **middle cardiac vein.** It is located on the posterior surface of the heart.

- The final 2 cardiac veins are also on the posterior surface of the heart:
- On the left posterior side is the **left marginal vein**.
- In the centre is the **left posterior ventricular vein** which runs along the posterior interventricular sulcus to join the coronary sinus.



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